LAB ASSIGNMENT – 3

JavaScript Programs | Deadline: 22 October 2026

How to run: Copy each program into a separate file ending in **.html**, then open it in a web browser.

1. Create a BMI Calculator Using JavaScript

Aim: To calculate BMI using height and weight and display the category. Formula: BMI = Weight (kg) / Height (m)²

```
<!DOCTYPE html>
<html>
<head>
    <title>BMI Calculator</title>
</head>
<body>
    <h2>BMI Calculator</h2>

    <input type="number" id="height" placeholder="Height in cm">
    <input type="number" id="weight" placeholder="Weight in kg">

    <button onclick="calculateBMI()">Calculate BMI</button>
    <p id="result"></p>

    <script>
        function calculateBMI() {
            let h = Number(document.getElementById("height").value) / 100;
            let w = Number(document.getElementById("weight").value);

            if (h <= 0 || w <= 0) {
                document.getElementById("result").innerHTML =
                    "Enter valid height and weight";
                return;
            }

            let bmi = w / (h * h);
            let category;

            if (bmi < 18.5)
                category = "Underweight";
            else if (bmi < 25)
                category = "Normal";
            else
                category = "Overweight";

            document.getElementById("result").innerHTML =
                "BMI: " + bmi.toFixed(2) + "<br>Category: " + category;
        }
    </script>
</body>
</html>
```

Output: Displays the calculated BMI and its category.

2. Create a Signup Page and Apply Form Validations

Aim: To create a signup form and validate the user's input.

```
<!DOCTYPE html>
<html>
<head>
    <title>Signup Page</title>
</head>
<body>
    <h2>Signup Form</h2>

    <form onsubmit="return validateForm()">
        <input type="text" id="name" placeholder="Enter Name"><br><br>
        <input type="email" id="email" placeholder="Enter Email"><br><br>
        <input type="password" id="password" placeholder="Enter Password"><br><br>
        <input type="password" id="confirm" placeholder="Confirm Password"><br><br>
        <button type="submit">Sign Up</button>
    </form>

    <p id="message"></p>

    <script>
        function validateForm() {
            let name = document.getElementById("name").value.trim();
            let email = document.getElementById("email").value.trim();
            let password = document.getElementById("password").value;
            let confirm = document.getElementById("confirm").value;
            let message = document.getElementById("message");

            if (name === "" || email === "" ||
                password === "" || confirm === "") {
                message.innerHTML = "Please fill all fields";
                return false;
            }

            if (password.length < 8) {
                message.innerHTML =
                    "Password must contain at least 8 characters";
                return false;
            }

            if (password !== confirm) {
                message.innerHTML = "Passwords do not match";
                return false;
            }

            message.innerHTML = "Signup successful";
            return false;
        }
    </script>
</body>
</html>
```

Output: Checks empty fields, password length, and matching passwords.

3. Create a Page in Which the User Can Change the Color Theme

Aim: To allow users to change between light and dark themes.

```
<!DOCTYPE html>
<html>
<head>
    <title>Color Theme Changer</title>

    <style>
        body {
            background-color: white;
            color: black;
            text-align: center;
            font-family: Arial;
        }

        .dark {
            background-color: black;
            color: white;
        }
    </style>
</head>
<body>
    <h1>Color Theme Changer</h1>
    <p>Choose your preferred theme.</p>

    <button onclick="changeTheme('light')">Light Theme</button>
    <button onclick="changeTheme('dark')">Dark Theme</button>

    <script>
        function changeTheme(theme) {
            if (theme === "dark") {
                document.body.classList.add("dark");
            } else {
                document.body.classList.remove("dark");
            }
        }
    </script>
</body>
</html>
```

Output: The buttons change the webpage between light and dark themes.

4. Create a Digital Clock Using JavaScript

Aim: To display the current time and update it every second.

```
<!DOCTYPE html>
<html>
<head>
    <title>Digital Clock</title>
</head>
<body>
    <h1>Digital Clock</h1>
    <h2 id="clock"></h2>

    <script>
        function showTime() {
            let time = new Date();
            document.getElementById("clock").innerHTML =
                time.toLocaleTimeString();
        }

        showTime();
        setInterval(showTime, 1000);
    </script>
</body>
</html>
```

Output: The current time appears and updates every second.

5. Create a To-Do List App

Aim: To add tasks, mark tasks as completed, and delete tasks.

```
<!DOCTYPE html>
<html>
<head>
    <title>To-Do List</title>
</head>
<body>
    <h1>To-Do List App</h1>

    <input type="text" id="task" placeholder="Enter a task">
    <button onclick="addTask()">Add Task</button>
    <ul id="taskList"></ul>

    <script>
        function addTask() {
            let input = document.getElementById("task");
            let task = input.value.trim();

            if (task === "") {
                alert("Please enter a task");
                return;
            }

            let li = document.createElement("li");
            let span = document.createElement("span");
            span.textContent = task;

            span.onclick = function() {
                if (span.style.textDecoration === "line-through") {
                    span.style.textDecoration = "none";
                } else {
                    span.style.textDecoration = "line-through";
                }
            };

            let button = document.createElement("button");
            button.textContent = "Delete";
            button.onclick = function() {
                li.remove();
            };

            li.appendChild(span);
            li.appendChild(document.createTextNode(" "));
            li.appendChild(button);
            document.getElementById("taskList").appendChild(li);
            input.value = "";
        }
    </script>
</body>
</html>
```

Output: Add tasks, click task text to mark completed, and click Delete to remove a task.